MARIAN GIRLS AND BOYS SECONDARY SCHOOLS
FORM ONE ANNUAL EXAMINATION, NOVEMBER 2012
BIOLOGY

TIME: 2HOURS **Monday 19th November 2012, 8.00pm**

INSTRUCTIONS

1. **Answer ALL questions**
2. **All answers must be written in a space provided for each question**

FOR EXAMINER'S USE ONLY

SECTIONS	A	B	C
MARKS			
TOTAL MARKS			

This paper consists of 6 printed pages.

SECTION A (50 marks)

1. Select the correct answer in each item and write its letter in the table provided below.

(i) If you are asked to prepare a first Aid Kit with its components, which of the following you will not include?

A. Iodine
B. Medicine to cure diseases
C. Pair of scissors
D. Sterilized cotton wool

(ii) Which of the following organizations show the correct order from simplest to the most complex ?
A. Cell, organ, tissue system and organism
B. Cell, tissue, organ, system and organism
C. Organ, cell, tissue, system and organism
D. Organ, tissue, cell, system and organism

(iii) An unexpected event resulting from carelessness, unawareness and ignorance is
A. Accident
B. Electric shock
C. Miracle
D. Bad lack

(iv) Which of the following is expected to happen if blood supply to the brain is reduced?
A. Fainting B. Shivering C. Shock D. Vomiting E. Sweating

(v) Which of the following is the chemical substance that catches fire easily.
A. Corrosive B. Flammable C. Harmful E. Irritant F. Oxidant

(vi) Which of the following is used to control all activities in a plant cell?
A. Cell wall B. Cell vacuole C. Chloroplast D. Cytoplasm F. Nucleus

(vii) All of the following are viral diseases:
A. Cancer, meningitis and yellow fever
B. Ringworm, meningitis and measles
C. Flu, measles and yellow fiver
D. Small pox, flu and goiter

(viii) The characteristics of living things are
A. Cell membrane, breathing, growth, feeding and locomotion.
B. Nucleus, irritability reproduction, growth and excretion
C. Reproduction, breathing, growth, excretion and irritability
D. Breathing growth, locomotion, irritability and cell wall

(ix) If you look at cells in frog muscles, you will observe that these cells contain nuclei but lack cell wall. The statement that frog muscle cells contain nuclei but no cell wall is termed as:
A. A hypothesis B. A. conclusion C. An assumption D. None of the above

(x) The following are the precautions to consider when handling a light microscope except.
A. Use two hands when carrying a light microscope
B. After use clean the microscope and store it in a safe place (cage)
C. Use host water to clean the microscope D. Do not touch the optical parts with bare fingers

Item no	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answer										

2. The following statements are either correct or not correct. Write True if the statement is correct or false if the statement is not correct (write the answers in the table provided below)
   (i) All living cells contain chloroplasts
   (ii) A microscope is an instrument that is used in scientific studies to magnify very small speamens
   (iii) Malaria is caused by a female anopheles mosquito
   (iv) Sharing of towels can spread ST is
   (v) Some laboratory accidents can be avoided by reading warning signs on chemical containers
   (vi) The best way to dispose sewage is by an incineration
   (vii) It is a good manner to greet people only if they are older than you
   (viii) Kwashiorkor is a non-communicable disease
   (ix) Typhoid and genital herpes are bacterial diseases
   (x) Rape is a type of sexual harassment
   (xi) Poor waste disposal may led to eruption of epidemic diseases
   (xii) Mature red blood cells have very large nuclei
   (xiii) All types of bacteria cause diseases in human beings
   (xiv) The nucleus is the basic unit of life
   (xv) Genetics is the study of inheritance and variations

Item No	i	ii	iii	iv	v	vi	vii	viii	ix	x	xi	xii	xiii	xiv	xv
Answer															

3. Match the items in list A with the responses in list B by writing the letter of the correct response beside the item number in list A in the table below.

LIST A	LIST B
(i) Sel –infected disease	A. Undergoes locomotion
(ii) The substance produced by the body to attack the invading pathogen	B. Undergoes movement of curvature
(iii) Graduated beaker	C. Ecology
(iv) Bean plant	D. Genetics
(v) The study of relationship between an organism and its environment	E. Plague
(vi) Many diseases and symptoms	F. Meningitis
(vii) Viral and bacterial diseases	G. Schistosomiasis
(viii) The vector involved in transanission of bilharziasis	H. Snails
(ix) The discoverer of cell	I. Schwann Theodore
(x) The organelle missing in plant cells	J. Robert Hooke
	K. Antigen
	L. Antibody
	M. Used to estimate volume of liquid
	N. Used to measure the accurate volume of the liquid
	O. Opportunistic diseases
	P. Syndrome
	Q. Centuiole
	R. Ribosomes
	S. Sickle cell anemia
	T. Alcoholism

Item no	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answer										

SECTION B (35 marks)

4. Draw a large diagram of a plant cell and label 6 parts that are visible by light microscope

5. Study the various animal cells, bellow then answer the questions that follow :

(a) For each of the cells numbered 1 to 4, state one key modification that has occurred in comparison to the generalized animal cell

1._____

_____

2._____

_____

3. _____

_____

4_____

_____

(b) Name and define the process leading to the modification of cells 1- 4
Name of the process:_____

Definition _____

_____

(c) Give 4 points to justify the importance of the process named in (b) above
(i) _____

_____

(ii) _____

_____

(iii) _____

_____

(iv) _____

_____

(d) The following are the descriptions of functions of cell organelles. For each of the description, name the organelle responsible
(i) Permeates the materials in and out of the cell selectively _____
(ii) It plays a role of producing energy thus nick named the power house _____
(iii) Double membranous structure carrying the chlorophyll molecules _____
(iv) Transports proteins synthesized on its surface._____
(v) Synthesizes proteins in the cell _____

6. The following are causative agents or key diagnostic symptoms of various diseases. For each of them state the diseases concerned.
(a) Frontal headache and abdominal pain _____
(b) Stool and vomitus resembling rice water _____
(c) Severe diarrhosea and there are traces of blood in faces_____
(d) Presence of blood in urine and bladder obstruction _____
(e) It is caused by a bacterium Pesturella pestis and spread by rat flea_____

SECTION C (15mark)

7. List various ways of classifying wastes citing are example in each. (15 marks)